



Toward Robust Machine Learning Models in Voice and Speech Science: A Hands-On Demonstration of Selection and Evaluation Biases

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House keeping

Link to access the materials:



You can run the code without installing any software on your computer

<https://colab.research.google.com>

ChatGPT/CoPilot

Two parts:

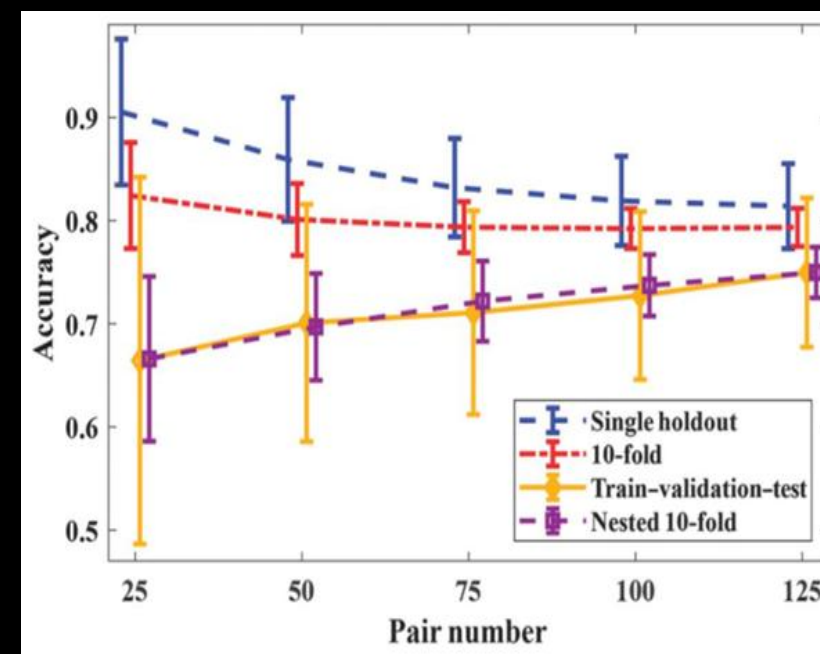
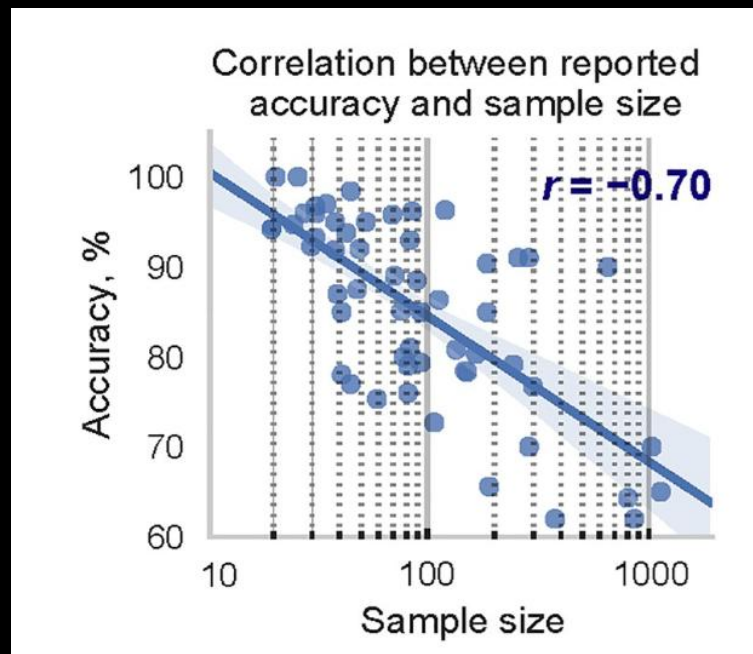
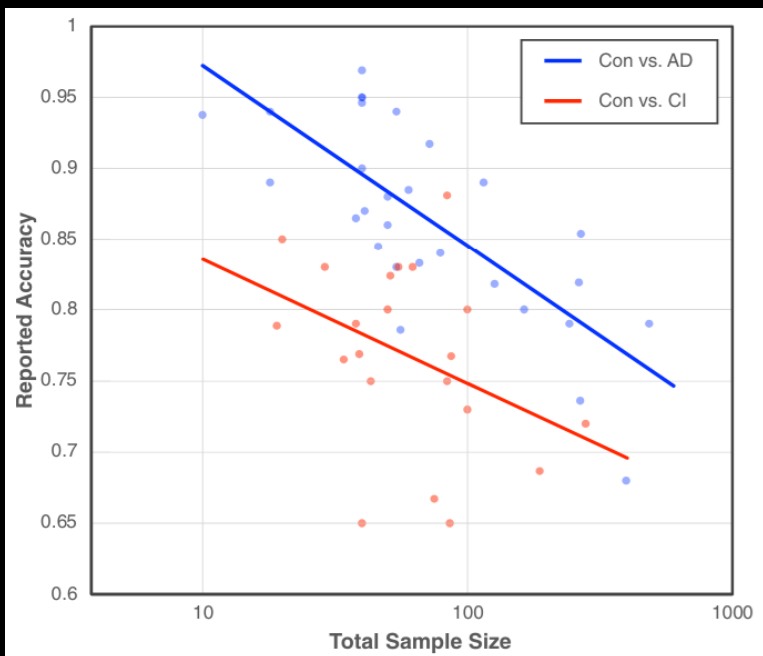
- Evaluation bias
- Selection bias

Supervised ML

- Ingredients:
 - Features (e.g., different voice measures)
 - Label (e.g., patients vs controls)
 - Algorithm(s)
- Complex data + exploring capability → Find “SOME” difference → Bias
- Implicit but important assumption
 - The groups are “SIMILAR” in all aspects except the target condition
- ML pipeline could be very complex and often includes multiple levels of optimization: feature selection, model selection, hyperparameter optimization, etc.
→ Inappropriate implementation → Bias



Generalizability of ML in health science



1: Berisha, V. et al. (2021). Digital medicine and the curse of dimensionality. NPJ digital medicine, 4(1), 153.

2: Vabalas, A. et al. (2019). Machine learning algorithm validation with a limited sample size. PloS one, 14(11), e0224365.

3: Ghasemzadeh, H. et al. (2024). Toward Generalizable Machine Learning Models in Speech, Language, and Hearing Sciences: Estimating Sample Size and Reducing Overfitting. JSLHR, 67(3), 753-781.

Outline

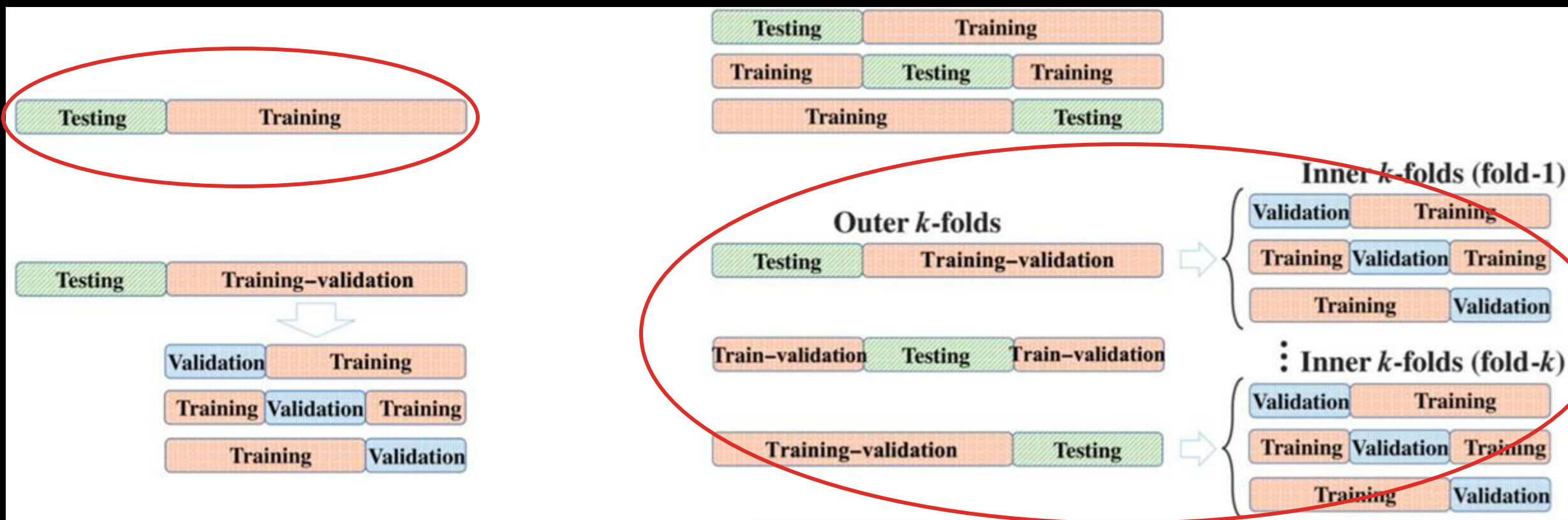
- Evaluation methodology
- Power analysis
- Imbalanced dataset

Module 1 – Evaluation Methodology ¹

¹ Ghasemzadeh, H. et al. (2024). Toward Generalizable Machine Learning Models in Speech, Language, and Hearing Sciences: Estimating Sample Size and Reducing Overfitting. JSLHR, 67(3), 753-781.

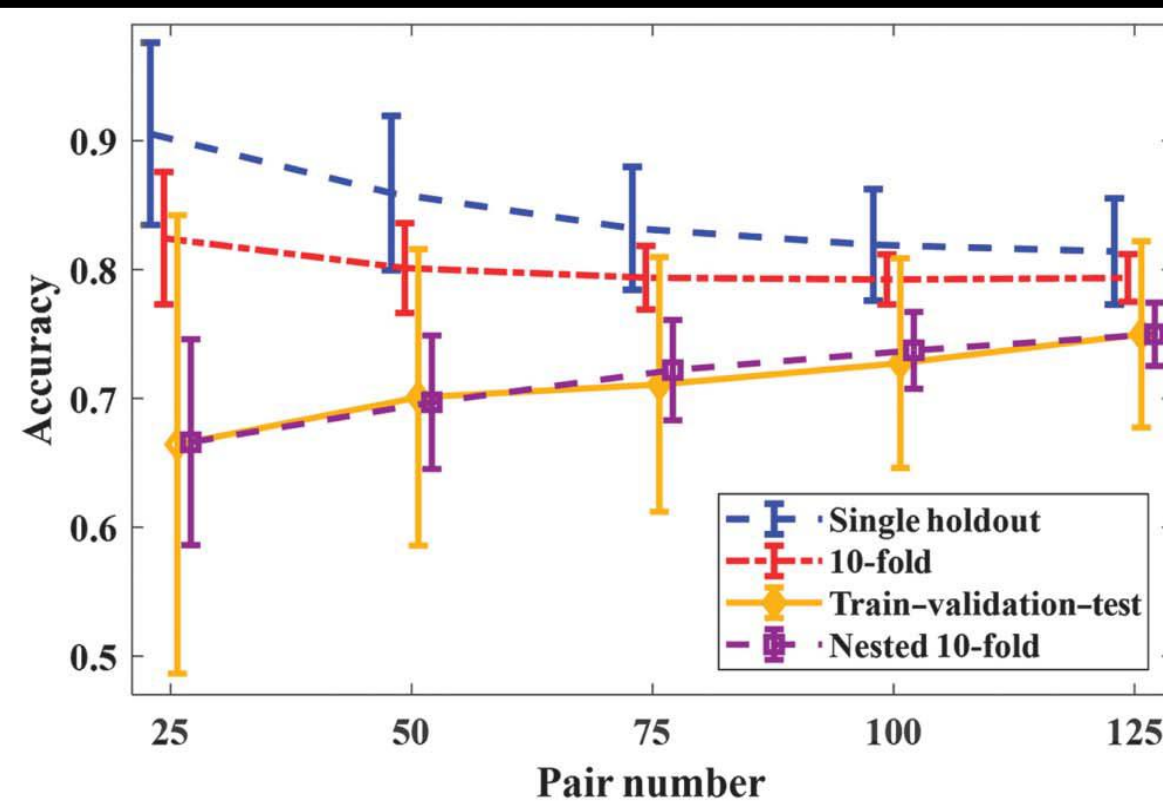
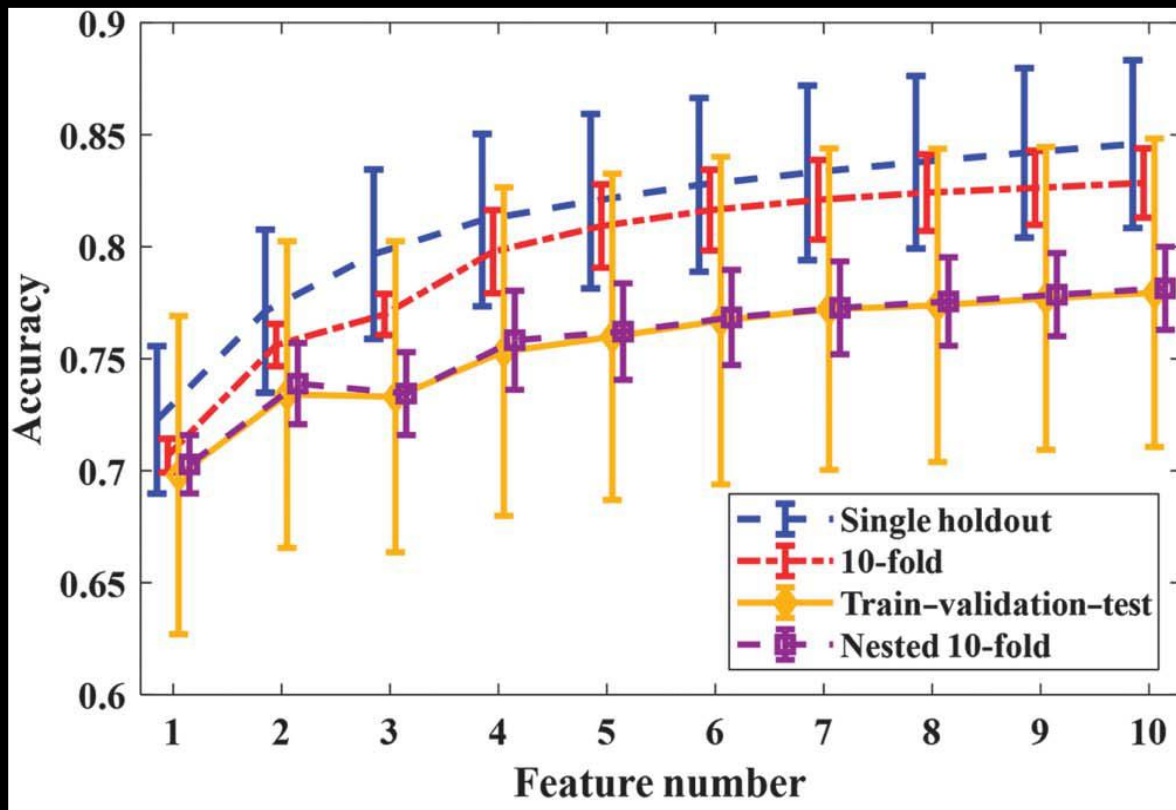
Cross-validation

Assumption: a model optimization component is present in the ML pipeline



Cross-validation

- Experimentation with real data (24 features, 136 pairs of participants)¹



¹ Ghasemzadeh, H. et al. (2024). Toward Generalizable Machine Learning Models in Speech, Language, and Hearing Sciences: Estimating Sample Size and Reducing Overfitting. JSLHR, 67(3), 753-781.

Cross-validation

Experimentation with synthesized data

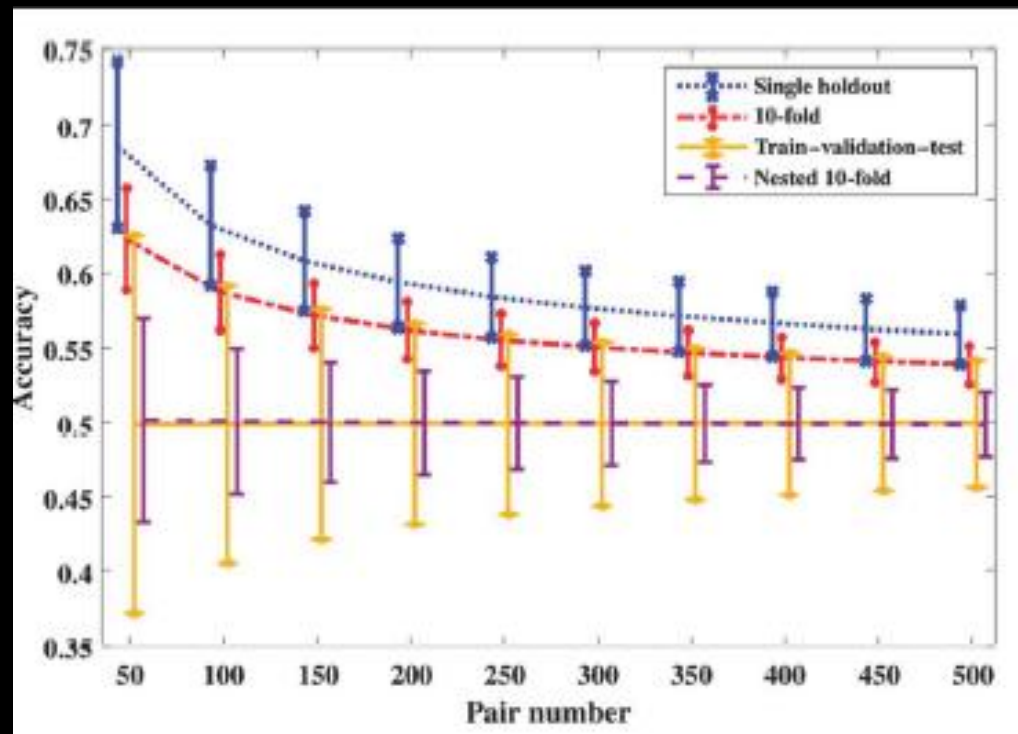
- Generating binary balanced datasets with different
 - number of features
 - sample sizes (n)
 - Systematically change the discriminative power of the features
 - Evaluate the effect of different cross validations

You can run it locally, on google Colab, or CoPilot/Chat GPT for free
<https://colab.research.google.com>

Demonstration



Cross-validation

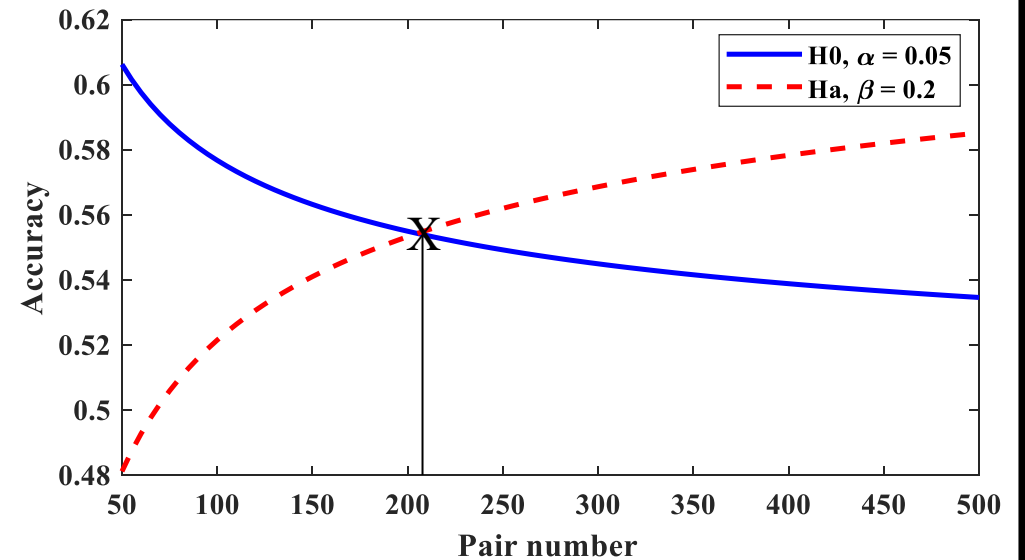
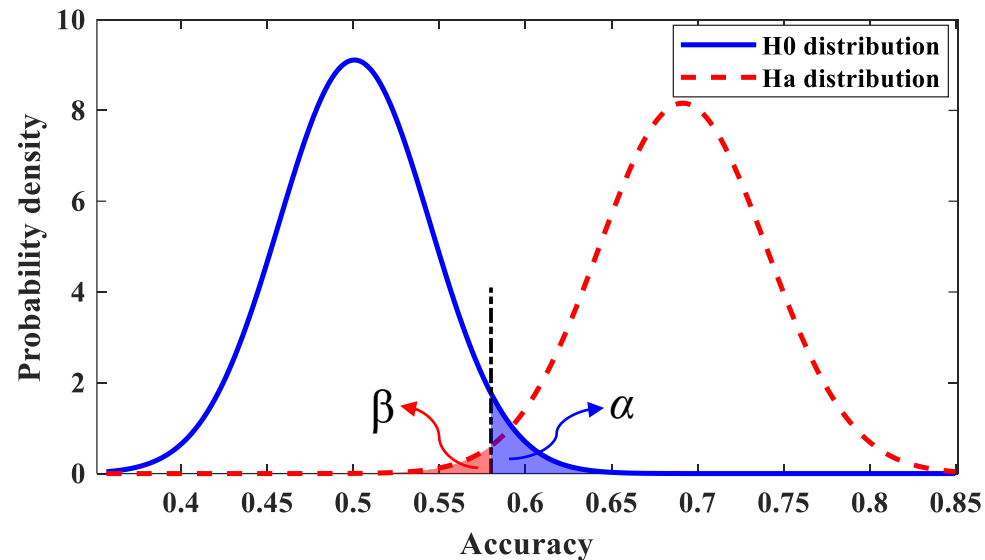


¹ Ghasemzadeh, H. et al. (2024). Toward Generalizable Machine Learning Models in Speech, Language, and Hearing Sciences: Estimating Sample Size and Reducing Overfitting. JSLHR, 67(3), 753-781.

Module 2 – Power Analysis ¹

¹ Ghasemzadeh, H. et al. (2024). Toward Generalizable Machine Learning Models in Speech, Language, and Hearing Sciences: Estimating Sample Size and Reducing Overfitting. JSLHR, 67(3), 753-781.

Power analysis

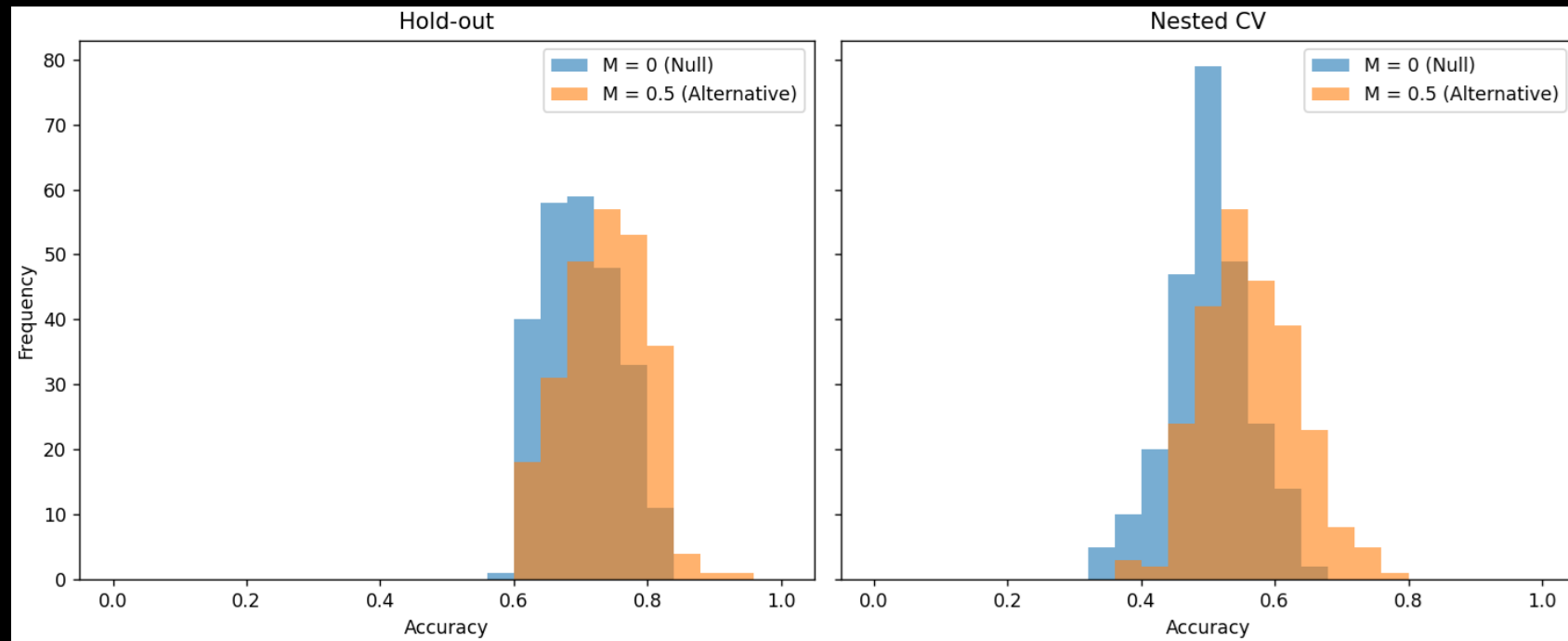


- α : probability of type I error
- β : probability of type II error ($1 - \beta = \text{power}$)

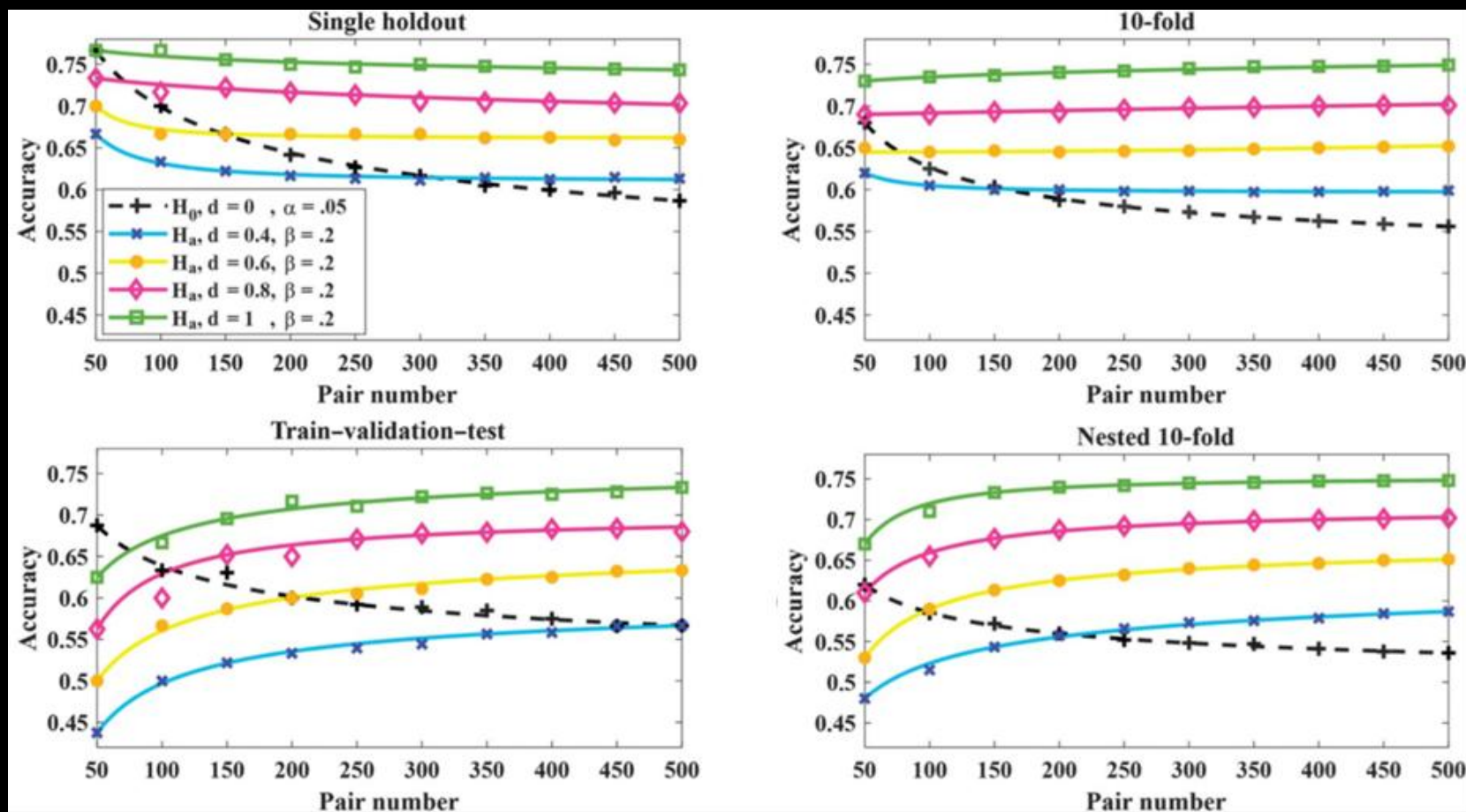
Demonstration



$n_{\text{reps}} = 250$



Power analysis

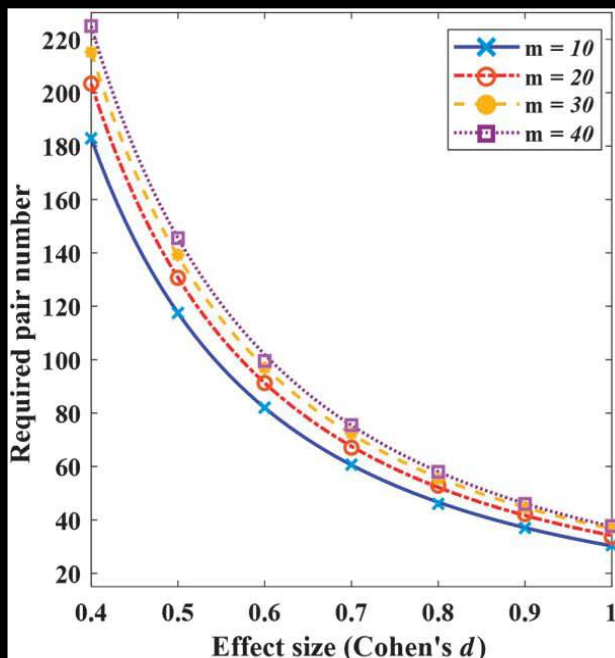


K-fold > nested k-fold > single hold out > train-validation-test

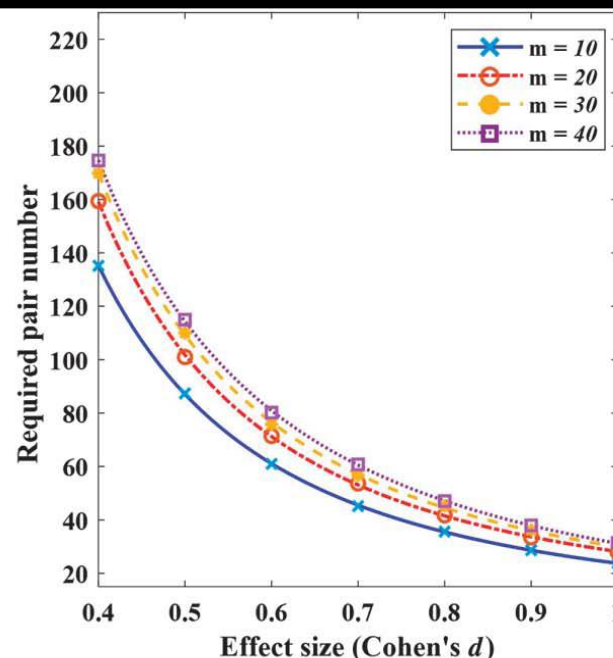
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Power analysis: nested k-fold

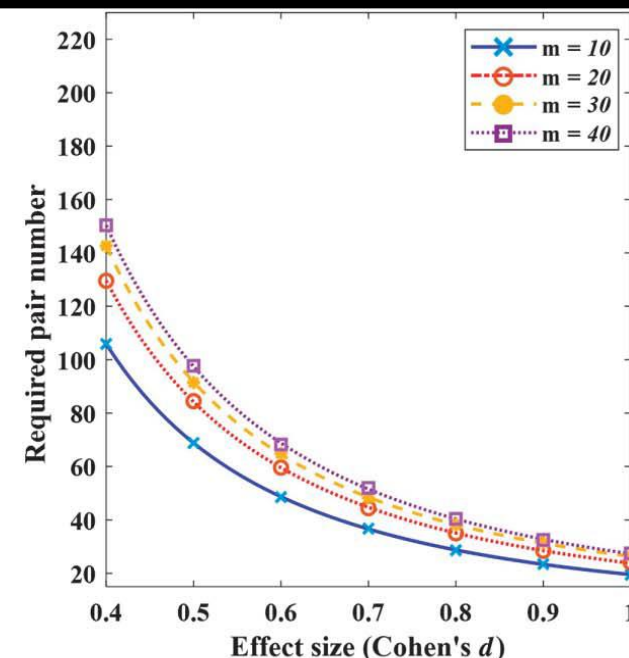
2 discriminative features



3 discriminative features



4 discriminative features



$$N_r = ad_0^b + c$$

$$a = 39.37 - 6.718l_0 + 0.263m_0$$

$$b = -1.985 - 0.023l_0 + 0.001m_0$$

$$c = -0.886 + 1.507l_0 - 0.015m_0$$

m_0 : Dimensionality of the feature space

l_0 : Number of discriminative (selected) features

d_0 : Cohen's d effect size of discriminative features



Module 3 – Imbalanced Dataset

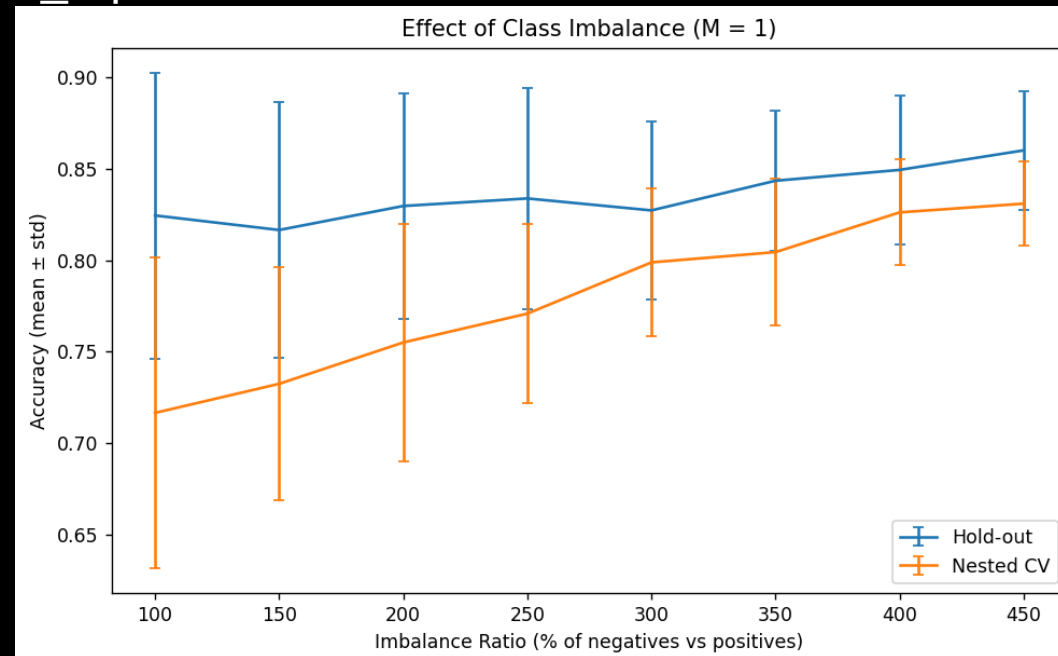


What is the baseline (random) performance for a ML model?

Demonstration



n_reps = 50



Thank you for your kind attention!

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